PQRH 0001: Quant And Reasoning Aptitude

Introduction: This subject has been designed keeping in mind the requirements of those students who intend to get good scores in placement drives of different companies and to prepare them to successfully achieve their goals.

Objective: *To help students develop the skills of simplifying their calculations and develop their skills to solve Problems or situations that involve logical thinking call for structure, for relationships between facts, and for chains of reasoning that "make sense."*

Credits: 04 **Semester I** **L–T–P-C: 0–0–4-2**

Module No.	Content	Teaching Hours (Approx.)
I	**Ratios and Proportion**: Defining Ratios, Properties of Ratios, Uses of Ratios Proportion: Defining Proportions, Continued Proportions, Uses of Proportions. **Variation and Partnership**: Defining Variation and it's types (Indirect and Direct Variations), Uses of Variations. **Average:** Average, Weighted average. Problems on ages, problems on distributions, one and two variable based questions. **Mixture and Alligation:** Alligation method, weighted avg. method, mixtures and replacement. **Coding and Decoding**: Coding based on rearrangement of letters, Coding based on replacement of letters, Opposite letters coding, Number coding, Symbol coding based on similarity, Fictitious language coding, Coding by comparison, Matrix Coding. **Ranking Test:** Sequential order of Arrangement, Potion Test, Time Sequence Test. **Doubt Session:** Doubt clearing session for students related to discussed topics.	12

II	**Time and Work**: Comparison of work force efficiencies. People working with same efficiency, People working with different efficiencies, Constructive & Distractive Work Concept. **Pipes & Cisterns:** Problems based on Pipes and Leakages. Comparison of capacities of pipes and cisterns. Calculating different times taken by different pipes with leakages to fill a given cistern. **Logarithms:** Introduction. Properties of Logarithms and Laws of Logarithms. Generalization. The Characteristics and Mantissa of a Logarithm. Base Change Rule. Results on Logarithmic Inequalities. **Surds & Indices:** Basics of arithmetic rules and operations. Simplification and comparison of given surds and indices etc. **Cube and Cuboid:** Cutting of a Cube and Cuboid, Counting of Blocks, Problem based on coloring the cube and cuboid with different Patterns and cutting them in small pieces. **Dice:** Finding Digits/Dots/Letters/Colors on the Opposite Face of Any Particular Face, Questions Related to Unfolded Dice. **Doubt Session:** Doubt clearing session for students related to discussed topics.	12

Reference Books/ Text Books / Cases:

- How to prepare for Quantitative aptitude test for CAT, by Arun Sharma, Fifth edition, T.M.H publication.
- How to Crack Test of Reasoning: In All Competitive Exam, by jaikishan and premkishan, Arihant publication.
- Analytical Reasoning Revised Edition by, M.K pandey, BSC publication.
- Quantitative aptitude by R.S aggarwal for Bank P.O/I.B.P.S P.O/A.A.O / Railways , S. chand publication.
- Patodi, B, Choudhary, A, Verbal Ability & Comprehension for CAT/ XAT/ IIFT/ CMAT/ MAT/ Bank PO/ SSC 2nd Edition, New Delhi, Disha Publication.
- Saxena, V, Bhatia, V, Crack WAT/GD/PI for MBA Admissions (Must for CAT/ XAT/ IIFT/ FMS/ SNAP/ NMAT/ CMAT Aspirants), New Delhi, Disha Publication.

Intended Outcome:

- Increase in ability to deduce inferences from problems.
- Enables a student to become agile in solving problems.
- Increase in logical and visualizing ability.
- Increase in perseverance to solve problems.